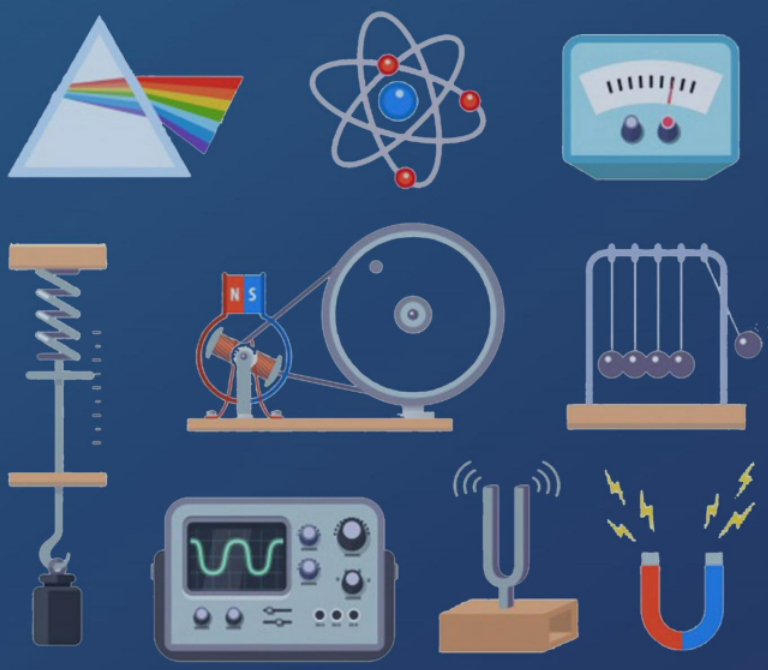
PHYSICS

Nuclear Physics



SHEET

eSaral.com

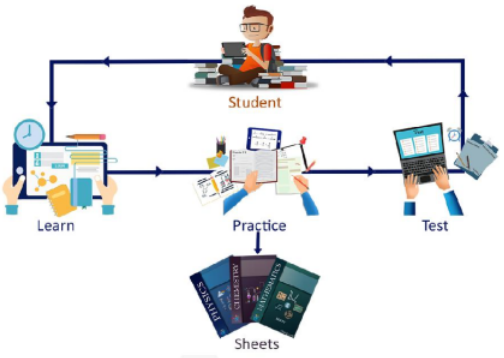
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 3
5.	Exercise 2	Page - 5
6.	Exercise 2A	Page - 6
7.	Answer Key	Page - 9

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Nuclear Physics	1,2			1,2
2	Nuclear force theory	3-5	1,11-14,16	1,2	
3	Binding energy and mass defect	6-18	2-6,15	3,4	8,10
4	Q value of a nuclear reaction	19-22	7-9	5-7	9,11,12
5	Where does the Q value go?				3
6	Which nucleus is more stable?	23	10		
7	Life lesson from Einstein				
8	Einstein and his world's most famous equation				
9	After Full Chapter	24		8,9	4-7

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Nuclear Physics

Q1. The mass number of a nucleus is -
(A) Always less than atomic number
(B) Always more than atomic number
(C) Equal to atomic number
(D) Sometimes more or equal to atomic number.

Q2. As the mass number A increases, which of the following quantities related to a nucleus do not change :
(A) mass
(B) volume
(C) density
(D) binding energy

Nuclear Force Theory

Q3. The surface area of a nucleus varies with mass number A as :
(A) $A^{2/3}$ (B) $A^{1/3}$
(C) A (D) None

Q4. The radius of the nuclues is proportional to, (if A is the atomic mass number).
(A) A (B) A^3
(C) $A^{1/3}$ (D) $A^{2/3}$

Q5. The nucleus of $_{56}Ba^{141}$ contains.
(A) 85 protons, 56 neutrons
(B) 55 protons ,86 neutrons
(C) 56 protons,85 neutrons
(D) 86 protons,55neutrons

Binding energy and mass defect

Q6. Masses of nucleus, neutron and protons are M, M_n and M_p respectivly, then :-
(A) $M = (A-Z)\,M_n + ZM_p$
(B) $M = ZM_n + (A-Z)\,M_p$
(C) $M > (A-Z)\,M_n + ZM_p$
(D) $M < (A-Z)\,M_n + ZM_p$

Q7. The value of binding energy for $_1H^2$, $_2He^4$, $_{26}Fe^{56}$, $_{92}U^{235}$ are 2.22, 28.3, 492 and 1786 units respectively. Then most stable nucleus is-
(A) $_{26}Fe^{56}$ (B) $_1H^2$
(C) $_{92}U^{235}$ (D) $_2He^4$

Q8. How much energy will be released when 10 kg of U^{235} gets completely converted into energy :-
(A) 5×10^{27} MeV
(B) 5×10^{24} MeV
(C) 9×10^{17} J
(D) All of these

Q9. If M_0 is the mass of an oxygen isotope $_8O^{17}$, M_p and M_n are the masses of a proton and a neutron, respectively, the nuclear binding energy of the isotope is -
(1) $(M_o - 8M_p)c^2$
(2) $(M_o - 8Mp - 9M_n)c^2$
(3) M_oc^2
(4) $(M_o - 17M_n)c^2$

Q10. The difference between mass of the nucleus and total mass of its constituents is called :
(A) Packing fraction
(B) Mass defect
(C) Binding energy
(D) Binding energy per neucleon

Q11. The parameter used to measure the stability of the nucleus is -

(A) Average binding energy

(B) No of protons

(C) No of neutrons

(D) No of electrons

Q12. When the number of nucleons in a nuclues increases the binding energy per nucleon.

(A) Increase continuously with mass number

(B) Decreases continuously with mass number

(C) Remains constant with mass number

(D) First increases and then decreases with increase in mass number

Q13. In a nuclear reaction some mass converts into energy. In this reaction total B.E of reactants when compared with that of products is -

(A) always greater

(B) always less

(C) either greater or less

(D) always equal

Q14. The particle A is converted to C via following reactions then -

$A \rightarrow B + {}_{2}He^{4}$

$B \rightarrow C + 2\,{}_{-1}e^{0}$

(A) A and C are isobars

(B) A and C are isotopes

(C) A and B are isobars

(D) A and B are isotopes

Q15. Energy released as mass of 2 amu is converted into energy is -

(A) 1.5×10^{-10} J

(B) 3×10^{-10} J

(C) 1863 J

(D) 931.5 Mev

Q16. The mass defect in a nucleus is 3.5 amu. Then the binding energy of the nucleus is -

(A) 32.58MeV

(B) 325.85 MeV

(C) 3260.25MeV

(D) 3.258 MeV

Q17. Let u denote one atomic mass unit. One atom of an element of mass number A has mass exactly equal to Au.

(A) for any value of A

(B) only for A = 1

(C) only for A = 12

(D) for any value of A provided the atom is stable

Q18. The binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.5 MeV. The energy required to remove a neutron from C^{13} is -

(A) 5.34 MeV (B) 5.5 MeV

(C) 9.5 MeV (D) 9.34 MeV

Q value of a nuclear reaction

Q19. Binding energy per nucleon of deutron is 1.112 MeV and binding energy per nucleon of an α-particle is 7.07 MeV, then in following process energy Q is :–

$$2({}_{1}H^{2}) \rightarrow {}_{2}He^{4} + Q$$

(A) 1 MeV (B) 11.9 MeV

(C) 23.8 MeV (D) 931 MeV

Q20. The nucleus of element X (A = 220) undergoes α-decay. If Q-value of the reaction is 5.5 MeV, then the kinetic energy of α-particle is :

(A) 5.4 MeV (B) 10.8 MeV

(C) 2.7 MeV (D) None

Q21. Consider the nuclear reaction :

$X^{200} \rightarrow A^{110} + B^{90} + \text{energy}$

The binding energy per nucleon for X^{200}, A^{110} and B^{90} is respectively 7.4MeV, 8.2MeV and 8.2Mev. The energy released is -

(A) 200MeV (B) 160MeV

(C) 110MeV (D) 90MeV

Q22. Binding energy per nucleon for deuteron and α Particle are x_1 and x_2 in the following equation :

${}_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + Q$

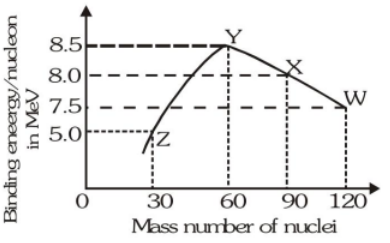
The value of energy (Q) released is :-

(A) $4(x_2 - x_1)$ (B) $4(x_1 + x_2)$

(C) $(x_1 - x_2)$ (D) $(x_1 + x_2)$

Which nucleus is more stable?

Q23. Binding energy per nucleon versus mass number curve for nuclei is shown in figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is :



(A) $Y \rightarrow 2Z$

(B) $W \rightarrow X + Z$

(C) $W \rightarrow 2Y$

(D) $X \rightarrow Y + Z$

After Full Chapter

Q24. A γ ray photon produces an electron positron pair, If the rest mass energy of electron is 0.51 Mev and the total kinetic energy of electron - positron pair is 0.78 MeV then the energy of γ - ray photon in Mev is :-

(A) 0.78 (B) 1.8

(C) 1.28 (D) 0.28

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If a nucleus is emitting e^- particle, its neutron to proton ratio $\left[\frac{n}{p}\right]$ will :–

(A) increase

(B) decrease

(C) remain unchanged

(D) can't be determined

Q2. If radius of the ${}^{27}_{13}Al$ nucleus is estimated to be 3.6 fermi, then the radius of ${}^{125}_{52}Te$ nucleus be nearly-

(A) 6 fermi (B) 8 fermi

(C) 4 fermi (D) 5 fermi

Q3. The binding energies of nuclei X and Y are E_1 and E_2 respectively. Two atoms of X fuse to give one atom of Y and an energy Q is released. Then:

(A) $Q = 2E_1 - E_2$

(B) $Q = E_2 - 2E_1$

(C) $Q = 2E_1 + E_2$

(D) $Q = 2E_2 + E_1$

Q4. The correct statements about binding energy is -

(A) It is the sum of the rest mass energies of nucleons minus the rest mass energy of the nucleus.

(B) It is the energy released when the nucleons combine to form a nucleus.

(C) It is the energy required to break a given nucleus into its constituent nucleons.

(D) It is the sum of the kinetic energies of all the nucleons in the nucleus.

Q5. The binding energies of a deuteron and an α-particle are 1.125, 7.2 MeV/nucleon respectively. The more stable of the two, is-

(A) deuteron

(B) α-particle

(C) both

(D) sometimes deutron and sometimes α-particle

Q6. Mass defect of an atom refers to -

(A) Inaccurate measurement of mass of neutrons

(B) Mass annihilated to produce energy to bind the nucleons

(C) Packing fraction

(D) Difference in the number of neutrons and protons in the nucleus

Q7. The binding energies of the atom of elements A & B are E_a & E_b respectively. Three atoms of the element B fuse to give one atom of element A. This fusion process is accompanied by release of energy e. Then E_a, E_b are related to each other as :

(A) $E_a + e = 3E_b$

(B) $E_a = 3E_b$

(C) $E_a - e = 3E_b$

(D) $E_a + 3E_b + e = 0$

Q8. The binding energy per nucleon of deuteron (^2_1H) and helium nucleus (^4_2He) is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is -

(A) 13.9 MeV (B) 26.9 MeV

(C) 23.6 MeV (D) 19.2 MeV

Q9. Which of these are conserved in a nucleus reaction :

(A) Mass number (B) Charge number

(C) Energy (D) None of these

Q10. The instability of the nucleus can be due to various causes. An unstable nucleus emits radiations if possible to transform into less unstable state. Then the cause and the result can be :

(A) a nucleus of excess nucleons is α active

(B) an excited nucleus of excess protons is β^- active

(C) an excited nucleus of excess protons is β^+ active

(D) an nucleus of excess neutrons is β^- active

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question. 11 to 13

Atomic nucleus is central core of every atom in which the whole of positive charge and almost entire mass of atom is concentrated. It is a tiny sphere of radius R is given by $R = R_o A^{1/3}$, where $R_o = 1.4 \times 10^{-15}$ m, a constant and A is the mass number of the nucleus.

Q11. A graph between '$\log\left(\frac{R}{R_o}\right)$, and 'log A'

(A) Is a parabola

(B) Is a straight line passing through origin

(C) Is a straight line have an intercept

(D) Is an ellipse

Q12. On increasing the value of 'A' the density of the nucleus.

(A) Increases

(B) Decreases

(C) Remains constant

(D) None

Q13. The radius of the nucleus of mass number 125 is -

(A) 175×10^{-15} m (B) 35×10^{-15} m

(C) 70×10^{-15} m (D) 7×10^{-15} m

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q14. Carbon has two stable isotopes. Natural carbon has 98.9% carbon–12 and 1.1% carbon–13, calculate the average atomic weight of carbon in amu.

Q15. Calculate the binding energy of $_{17}Cl^{35}$ (in MeV)if mass of $_{17}Cl^{35}$ nucleus is 34.98 amu, mass of neutron is 1.008665 amu, and mass of proton is 1.007277 amu.

SUBJECTIVE TYPE

Q16. Two nuclei have their mass numbers in ratio 1 : 3. What is the ratio of nuclear densities?

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Let F_{pp}, F_{pn} and F_{nn} denote the nuclear force between proton–proton, proton–neutron and neutron–neutron pair respectively. When separation is 1 fm :-

(A) $F_{pp} < F_{pn} = F_{nn}$ (B) $F_{pp} > F_{pn} = F_{nn}$

(C) $F_{pp} = F_{pn} = F_{nn}$ (D) $F_{pp} < F_{pn} < F_{nn}$

Q2. A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2 : 1. The ratio of their nuclear sizes will be :-

(A) $2^{1/3} : 1$ (B) $1 : 3^{1/2}$

(C) $3^{1/2} : 1$ (D) $1 : 2^{1/3}$

Q3. If a star can convert all the He nuclei completely into oxygen nuclei. The energy released per oxygen nuclei is : [Mass of the helium nucleus is 4.0026 amu and mass of oxygen nucleus is 15.9994 amu]

(A) 7.6 MeV

(B) 56.12 MeV

(C) 10.24 MeV

(D) 23.4 MeV

Q4. The mass of proton is 1.0073 u and that of neutron is 1.0087 u (u = atomic mass unit).

The binding energy of $_2^4He$ is :-

[Given helium nucleus mass $\approx$4.0015u]

(A) 0.0305 J

(B) 0.0305 erg

(C) 28.4 MeV

(D) 0.061 u

Q5. The binding energy of deuteron is 2.2 MeV and that of $_2^4$He is 28 MeV. If two deuterons are fused to form one ^4_2He then the energy released is :-

(A) 23.6 MeV (B) 19.2 MeV

(C) 30.2 MeV (D) 25.8 MeV

Q6. Per nucleon energy of $_3Li^7$ and $_2He^4$ nucleus is 5.60 MeV and 7.06 MeV, then in $_3Li^7 + {}_1P^1 \rightarrow 2{}_2He^4$ energy released is :-

(A) 29.6 MeV (B) 2.4 MeV

(C) 8.4 MeV (D) 17.3 MeV

Q7. In an α-decay the Kinetic energy of α particle is 48 MeV and Q-value of the reaction is 50 MeV. The mass number of the mother nucleus is : (Assume that daughter nucleus is in ground state)

(A) 96 (B) 100

(C) 104 (D) none of these

Q8. When U^{238} nucleus originally at rest, decays by emitting an alpha particle having a speed u, the recoil speed of the residual nucleus is-

(A) $\frac{4u}{238}$ (B) $-\frac{4u}{234}$

(C) $\frac{4u}{234}$ (D) $-\frac{4u}{238}$

Q9. An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to -

(1) v^2 (2) 1/m

(3) $1/v^4$ (4) 1/Ze

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Consider the following nuclear reactions and select the correct statements from the options that follow :

Reaction I : $n \rightarrow p + e^- + \bar{\nu}$

Reaction II : $p \rightarrow n + e^+ + \nu$

(A) Free neutron is unstable, therefore reaction I is possible

(B) Free proton is stable, therefore reaction II is not possible

(C) Inside a nucleus, both decays (reaction I and II) are possible

(D) Inside a nucleus, reaction I is not possible but reaction II is possible.

Q2. A nuclear transformation is denoted by $X(n, \alpha) \rightarrow {}^7_3Li$. Which of the following is the nucleus of element X ?

(A) $^{12}_6C$ (B) $^{10}_5B$

(C) ^9_5B (D) $^{11}_4Be$

Q3. In the options given below, let E denote the rest mass energy of a nucleus and *n* a neutron. The correct option is :-

(A) $E\left(^{236}_{92}U\right) > E\left(^{137}_{53}I\right) + E\left(^{97}_{39}Y\right) + 2E(n)$

(B) $E\left(^{236}_{92}U\right) < E\left(^{137}_{53}I\right) + E\left(^{97}_{39}Y\right) + 2E(n)$

(C) $E\left(^{236}_{92}U\right) < E\left(^{140}_{56}Ba\right) + E\left(^{94}_{36}Kr\right) + 2E(n)$

(D) $E\left(^{236}_{92}U\right) = E\left(^{140}_{56}Ba\right) + E\left(^{94}_{36}Kr\right) + 2E(n)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 4 to 7

When subatomic particles undergo reactions, energy is conserved, but mass is not necessarily conserved. However, a particle's mass 'contributes' to its total energy, in accordance with Einstein's famous equation, $E = mc^2$

In this equation, E denotes the equivalent energy when a particle of mass m is converted into energy. The particle can also have additional energy due to its motion and its interactions with other particles.

Consider a neutron at rest, and well separated from other particles. It decays into a proton, an electron, and an undetected third particle:

Neutron $\rightarrow$ proton + electron + third particle

The table below summarizes some data from a single nuetron decay. Column 2 shows the rest mass of the particle times the speed of light squared.

Particle	**Mass x c^2 (MeV)**	**Kinetic Energy (MeV)**
Neutron	940.97	0
Proton	939.66	0.02
Electron	0.51	0.42

Q4. Assuming the table contains no major errors, what can we conclude about the (mass x c^2) of the undetected third particle?

(A) It is 0.90 MeV

(B) It is 0.36 MeV

(C) It is less than or equal to 0.90 MeV; but we cannot be more precise

(D) It is less than or equal to 0.36 MeV; but we cannot be more precise

Q5. From the given table, which properties of the undetected third particle can we calculate?

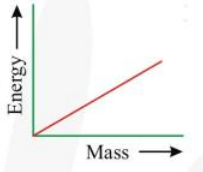
(A) Total energy, but not kinetic energy

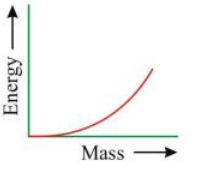
(B) Kinetic energy, but not total energy

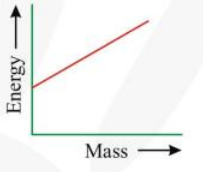
(C) Both total energy and kinetic energy

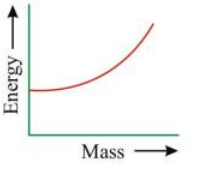
(D) Neither total energy nor kinetic energy

Q6. Consider an ensemble of particles, all of which have the same positive kinetic energy but different masses. For this ensemble, which graph best represents the relationship between the particle's mass and its total energy?

(A) 

(B) 

(C) 

(D) 

Q7. Could this reaction occur?

Proton $\rightarrow$ neutron + other particles

(A) Yes, if the other particles have much more kinetic energy than mass energy

(B) Yes, but only if the proton has potential energy (due to interactons with other particles)

(C) No, because a neutron is more massive than a proton

(D) No, because a proton is positively charged while a neutron is electrically neutral.

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q8. In the following, column I lists some physical quantities & the column II gives approx. energy values associated with some of them. Choose the appropriate value of energy from column II for each of the physical quantities in column I and write the corresponding letter A, B, C etc. against the number (i), (ii), (iii), etc. of the physical quantity in the answer book. In your answer, the sequence of column I should be maintained.

Column I	**Column II**
(i) Energy of thermal neutrons	(A) 0.025 eV
(ii) Energy of X-rays	(B) 0.5 eV
(iii) Binding energy per nucleon	(C) 3 eV
(iv) Photoelectric threshold of metal	(D) 20 eV
	(E) 10 keV
	(F) 8 MeV

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q9. The binding energies per nucleon for deuteron ($_1H^2$) and helium ($_2He^4$) are 1.1 MeV and 7.0 MeV respectively. Find the energy released (in MeV) when two deuterons fuse to form a helium nucleus ($_2He^4$).

SUBJECTIVE TYPE

Q10. The Q value of a nuclear reaction $A + b \rightarrow C + d$ is defined by

$Q = [m_A + m_b - m_C - m_d]c^2$

where the masses refer to the respective nuclei. Determine from the given data the Q-value of the following reactions and state whether the reactions are exothermic or endothermic.

(i) $_1^1H + _1^3H \rightarrow _1^2H + _1^2H$

(ii) $_6^{12}C + _6^{12}C \rightarrow _{10}^{20}Ne + _2^4He$

Atomic masses are given to be :

$m(_1^2H) = 2.014102$ u

$m(_1^3H) = 3.016049$ u

$m(_6^{12}C) = 12.000000$ u

$m(_{10}^{20}Ne) = 19.992439$ u

Q11. Suppose that the Sun consists entirely of hydrogen atom and releases the energy by the nuclear reaction, $4\,_1^1H \longrightarrow {}_2^4He$ with 26 MeV of energy released. If the total output power of the Sun is assumed to remain constant at 3.9×10^{26} W, find the time it will take to burn all the hydrogen. Take the mass of the Sun as 1.6×10^{30} kg.

Q12. The kinetic energy of an α – particle which flies out of the nucleus of a Ra^{226} atom in radioactive disintegration is 4.78 MeV. Find the total energy evolved during the escape of the α – particle.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	C	A	C	C	D	A	C	B	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	D	B	B	B	C	C	A	C	A
21	**22**	**23**	**24**						
B	A	C	B						

Ex-1A

1. B **2.** A **3.** B **4.** A,B,C **5.** B

6. B **7.** C **8.** C **9.** A,B,C **10.** A,C,D

11. B **12.** C **13.** D **14.** 12.011 u **15.** 278.92 MeV

16. 1:1

Ex-2								
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
C	D	C	C	A	D	B	C	B

Ex-2A

1. A,B,C **2.** B **3.** A **4.** D **5.** A

6. C **7.** B **8.** (i) A, (ii) E (iii) F (iv) C **9.** 23.6

10. (i) –4.054 MeV, endothermic (ii) 5.64 MeV, exothermic

11. $8/3\times10^{18}$ sec **12.** 4.87 MeV

-: Important Notes & Formulas :-